Answers to Supplementary Questions

Public Accountability and Works Committee

Inquiry into Data Centres

Protocol Policy Lab

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Chapter 1

Introduction and Approach

This document provides Protocol Policy Lab's answers to the two supplementary questions issued following the hearing of 1 May 2026. It is lodged as a standalone response. The inquiry to which it contributes is itself a novel exercise, described by its proponents as a process to "establish a parliamentary inquiry into data centre developments in NSW, the first of its kind in Australia" (The Greens NSW, 2026). That novelty matters for both questions, because it means New South Wales is forming its evidentiary and regulatory baseline at the same moment that the sector is expanding fastest.

1.1 Scope and the questions answered

The first supplementary question asks whether Protocol holds data on the typical Power Usage Effectiveness (PUE) of data centres currently operating in New South Wales, whether such information is publicly available, and whether Protocol supports mandatory PUE reporting as a baseline transparency measure. The second asks whether Protocol is aware of evidence or analysis bearing on whether energy gentrification is occurring or likely to occur in New South Wales, with particular reference to Western Sydney, where data centre clustering is most concentrated. Chapter 2 answers the first and Chapter 3 the second. Chapter 4 draws out the single thread that connects them, which is that neither efficiency policy nor consumer protection can operate ahead of measurement and disclosure.

1.2 Evidence base and method

The answers draw on two bodies of evidence. The first is the peer-reviewed and technical literature on data centre energy, water and grid integration. The second is the primary and grey literature specific to the Australian National Electricity Market and to New South Wales, including the Australian Energy Market Operator forecasts as relayed in the New South Wales Government's own consultation paper, transmission planning by Transgrid, and analysis by Energy Consumers Australia and the United States Studies Centre. Quotations throughout are reproduced verbatim from the cited source. Measured quantities are distinguished from projections, and projections are attributed to their modelling scenario wherever the source specifies one. Where no New South Wales figure exists, this is stated rather than inferred from overseas data.

1.3 Power Usage Effectiveness and the limits of a single metric

Power Usage Effectiveness is the ratio of a facility's total energy to the energy delivered to its information technology equipment. A value of 1.0 is the theoretical ideal in which no energy is spent on overheads, and lower values denote greater efficiency. The metric is mature and standardised, having been established such that "the assessment method of Power Usage Effectiveness has become an international standard for measuring energy efficiency and is widely used in the design and operation of data centers" (Uanov and Begimbetova, 2022). Cooling is the dominant overhead the ratio captures, since "the energy consumption of the cooling system for the machine room accounts for a significant part of the operating costs of the building" (Uanov and Begimbetova, 2022).

PUE is a necessary instrument but an insufficient one, and this submission is careful not to treat it as a complete measure. It is a ratio, so it is silent on absolute consumption, on the carbon intensity of the electricity drawn, and on water. The water dimension is consequential and under-measured, given that "Data centres consume water directly for cooling, in some cases 57% sourced from potable water" (Mytton, 2021), and that the academic literature records how, for data centres, "their water footprint (WF) has so far received little to no attention" (Ristic, Madani, and Makuch, 2015). Water intensity is also enormously site-dependent, with one assessment finding "workload-level water use variations exceeding 10,000-fold" (Lei et al., 2025). A PUE figure can improve while absolute energy, emissions and water all rise, which is why the recent literature advocates "mandatory resource transparency, efficiency regulation, and lifecycle accountability" rather than a single efficiency ratio (Iman, 2025).

The insufficiency is compounded by rebound. Efficiency gains at the facility level are routinely offset by growth in demand, so that "the energy-intensive nature of digital infrastructure may conversely increase emissions", with the gains from efficiency repeatedly overtaken by rising consumption (Liu and Wang, 2025). The United States experience illustrates the same point in reverse, where for two decades "comparable increases in electricity demand have been avoided through the adoption of key energy efficiency measures and a shift towards large cloud-based service providers" (Shehabi et al., 2018), a stabilisation that the artificial intelligence load is now ending. The practical implication for New South Wales is that an efficiency-conditioned regime should require a basket of measured indicators, being PUE alongside water usage effectiveness, absolute energy and renewable share, not PUE in isolation.

1.4 The standard of inference adopted

Two of the conclusions in this submission rest on inference rather than direct local measurement, and the standard applied to them is stated plainly here so that the Committee can weigh them appropriately. For the question of mandatory reporting, the argument is that measurement to a common standard and disclosure are a precondition for any efficiency-conditioned or capacity-allocation policy, since a standard that is not measured cannot be enforced and a standard that is not disclosed cannot be scrutinised. For the question of energy gentrification, the argument is deliberately bounded. Protocol does not assert that energy gentrification is occurring in New South Wales, nor that it will occur. The claim is the weaker and more defensible one that the conditions which preceded the effect in comparable jurisdictions are present in Western Sydney, that the outcome is therefore a material and foreseeable risk, and that because the outcome is contingent on policy settings it is preventable. This framing is adopted precisely because the contingency strengthens rather than weakens the case for early action, since a preventable harm warrants pre-emptive safeguards while it remains preventable.

Chapter 2

Supplementary Question 1: Power Usage Effectiveness Reporting

2.1 The question

> *Your submission discusses Singapore's managed capacity allocation regime, where data centre approvals are conditional on meeting a Power Usage Effectiveness (PUE) of 1.25 or better. Do you have data on the typical PUE of data centres currently operating in NSW? Is this information publicly available, and if not, do you support mandatory PUE reporting as a baseline transparency measure?*

In short, Protocol does not hold, and is not aware of, any comprehensive and publicly available dataset reporting the Power Usage Effectiveness of data centres operating in New South Wales. The information is not public because the only fit-for-purpose New South Wales instrument is voluntary and its underlying data are confidential. Protocol supports mandatory reporting as a baseline transparency measure, subject to the qualification that the obligation should capture a basket of measured indicators rather than Power Usage Effectiveness alone. The remainder of this chapter sets out the basis for each of these positions.

2.2 What is known about efficiency in operating data centres

The publicly defensible statements about typical Power Usage Effectiveness are international and national averages, not facility-level New South Wales figures. The most recent global industry survey places the worldwide average at roughly 1.5 and records that it has stagnated rather than improved across recent years (Uptime Institute, 2025). Commentary on the Australian fleet places the national average near 1.44 for 2023, a figure that itself derives from vendor analysis rather than any official register (Nlyte, 2024). These averages conceal a wide distribution. The most efficient hyperscale operators report fleet-wide values close to 1.1, and they can do so credibly only because they measure continuously, explaining that "Our calculations are based on continuously measuring the entire worldwide fleet performance of our data centers throughout the year" (Google, 2025). Singapore's condition of a Power Usage Effectiveness of 1.25 or better therefore sits below the Australian average and near the upper boundary of best practice, which is to say it is demanding but demonstrably achievable.

These figures should be read against the trajectory of demand that makes efficiency salient. Artificial intelligence has ended the long plateau in which efficiency gains offset workload growth, and data centres "have become one of the fastest-growing electricity consumers globally" (Sheng et al., 2026), with the underlying compute exhibiting "a Deep Learning Era compute doubling time of around 6 months" (Sevilla et al., 2022). The resulting consumption is also systematically understated by simple proxies, since "naive energy estimates based on FLOPs or theoretical GPU utilization significantly underestimate real-world energy consumption" (Fernandez et al., 2025). Reliable measurement at the facility, rather than estimation from first principles, is therefore the only sound basis for policy.

2.3 Why no New South Wales figure is available

The absence of a New South Wales figure is structural rather than incidental. New South Wales has access to a purpose-built instrument, the NABERS Energy for Data Centres scheme, which rates facilities from one to six stars on measured operational performance across three rating types, being Infrastructure, Information Technology Equipment, and Whole Facility (NABERS, 2024a). Participation in that scheme is voluntary. The scheme maintains a public register of certified ratings, but its coverage is partial because only operators who choose to participate and to publish appear in it, and the underlying energy data remain confidential (NABERS, 2024b). The aggregate efficiency of the New South Wales fleet is consequently unobservable.

The mandatory reporting that does exist is narrow. From 1 July 2025 the Commonwealth requires a minimum five-star NABERS Energy rating for data centres hosting Australian Government workloads, an obligation that binds only that subset of facilities (Law Society Journal, 2024). The National Greenhouse and Energy Reporting scheme captures total energy and emissions above a facility threshold of 100 terajoules but does not measure or require Power Usage Effectiveness, because it is an emissions accounting framework rather than an efficiency standard (Department of Climate Change, Energy, the Environment and Water, 2025). Proposed expansion of the Commercial Building Disclosure programme to data centres is the most likely future vehicle, advanced on the basis that "The commercial building sector is responsible for around 25% of electricity use and 10% of total emissions in Australia", but it remains in a consultation and roadmap phase and offers no near-term remedy (Department of Climate Change, Energy, the Environment and Water, 2024). The net position is that private operators outside the Commonwealth perimeter face no obligation to disclose efficiency at all.

2.4 International precedent for mandatory disclosure

The feasibility of mandatory disclosure is settled by precedent. The European Union has moved from voluntary benchmarking to a statutory regime requiring "monitoring and reporting of the energy performance of data centres", implemented through a "Delegated Regulation on the establishment of a common Union rating scheme for data centres (EU/2024/1364)" (European Commission, 2024). That regime is explicitly a basket rather than a single ratio, since the directive "requires data centers to report operational efficiency metrics such as power usage effectiveness (PUE) and water usage effectiveness (WUE), and adopt measures that optimize electricity and water usage" (Data Center Knowledge, 2026). The contrast with the United States is instructive, where "In the US, no federal equivalent of the EED exists, but state-level activity is picking up" (Data Center Knowledge, 2026). The emerging state instruments couple disclosure to network management, as with the Oregon measure that "establishes a special electricity rate for data centers and other large power consumers, incentivizing efficiency and grid-friendly load profiles" (Data Center Knowledge, 2026).

There is an instructive parallel in the governance of computing more generally, where reporting is treated as the precondition for any further lever. Compute is considered "a particularly effective point of intervention" precisely because "it is detectable, excludable, and quantifiable" (Sastry et al., 2024), and the frontier-regulation literature places "registration and reporting requirements to provide regulators with visibility into frontier AI development processes" ahead of substantive obligations (Anderljung et al., 2023). The same logic recommends that an efficiency standard for data centres begin with measurement and disclosure, not with an unenforceable target.

2.5 Why disclosure is necessary but not sufficient

Two qualifications discipline the recommendation. First, disclosure is necessary for accountability but not sufficient for outcomes, and it is useful to separate the two functions that transparency performs. The distinction in the governance literature "between principles of 'fishbowl transparency' and 'reasoned transparency'" captures the point, since publishing raw ratings serves public scrutiny while reporting to the regulator serves enforcement, and a well-designed regime needs both (Coglianese and Lehr, 2019). An efficiency-conditioned regime in the manner of Singapore does not strictly require a public register in order to function, because measurement reported to and verified by the regulator would suffice for enforcement; public disclosure is justified separately, as the layer that allows the standard and its administration to be scrutinised.

Second, Power Usage Effectiveness alone is the wrong target. It is gameable, since a facility can improve the ratio by raising its information technology baseline, and it is blind to absolute consumption, carbon and water. The literature accordingly advocates "transparency mechanisms, such as disclosing the GHG footprint of AI systems", situating efficiency within a broader resource and emissions frame in which "The ICT sector contributes up to 3.9 percent of global greenhouse gas (GHG) emissions-more than global air travel at 2.5 percent" (Hacker, 2024). The case for disclosing emissions specifically is strengthened by evidence that estimates omitted from primary reporting are wildly unreliable, with one analysis finding that "Estimates of emissions in papers that omitted them have been off 100x-100,000x" (Patterson et al., 2022). Water reinforces the same conclusion, since the most consequential figures have historically been concealed, as when "training the GPT-3 language model in Microsoft's state-of-the-art U.S. data centers can directly evaporate 700,000 liters of clean freshwater, but such information has been kept a secret" (Li et al., 2025). A mandate confined to Power Usage Effectiveness would license precisely the partial visibility that has allowed these costs to go unmeasured.

2.6 Recommendation

Protocol recommends that New South Wales make disclosure a condition of development consent and of continued operation for data centres above a defined information technology power threshold, calibrated on the European Union's 500 kilowatt model. The disclosed basket should comprise the NABERS Whole Facility rating and its underlying Power Usage Effectiveness, water usage effectiveness, absolute annual energy, and renewable share, reported annually to the regulator and published in aggregate and at facility level. NABERS already supplies the administrative spine and the measurement methodology, so the missing element is solely the obligation to measure and disclose. This sequencing is the precondition for every downstream lever the Committee may later consider, including a Singapore-style allocation of capacity against an efficiency standard, because capacity cannot be allocated against a benchmark that is neither measured nor published, and a target that cannot be audited cannot be enforced (Iman, 2025; Hacker, 2024).

Chapter 3

Supplementary Question 2: Energy Gentrification in Western Sydney

3.1 The question

> *You discussed the concept of "energy gentrification" – where concentrated data centre development drives up energy costs for local consumers – citing examples from the United States and Ireland. Are you aware of any evidence or analysis of whether this effect is occurring or likely to occur in NSW, particularly in Western Sydney where data centre clustering is most concentrated?*

Protocol is not aware of any published, New South Wales specific quantitative study that calculates an energy gentrification effect, being a measurable increase in residential energy costs attributable to data centre concentration, for the State or for Western Sydney. This is stated plainly at the outset. What follows is the evidence that does exist, which establishes the concept, documents the effect in comparable jurisdictions, identifies the causal mechanism, and shows that the conditions which produced the effect elsewhere are present in Western Sydney. The conclusion drawn is bounded accordingly. The effect is a material, foreseeable and preventable risk rather than a demonstrated fact, and it is the preventability that makes early action worthwhile.

3.2 The concept and its scholarly basis

Energy gentrification belongs to a wider scholarship on digital extractivism that treats the data centre as an extractive industry rather than a neutral utility. The framing is direct, holding that "AI is a technology of extraction: from the energy and minerals needed to build and sustain its infrastructure, to the exploited workers behind 'automated' services, to the data AI collects from us" (Crawford, 2021). Applied to electricity, this becomes the proposition that "The global data center industry relies on what this article defines as 'climate extraction'" (Brodie, 2020). The Irish case studies are the most developed, and they show how siting decisions are presented as benign while their fiscal and infrastructural logic is obscured, given that "Ireland has been advertised to and by data center developers because of its 'cool' climate while downplaying the importance of its low corporate tax rate" (Brodie, 2020). The cost of that arrangement is socialised, because the new infrastructures are "entangled with non-renewable energy systems and tax evasive capital, and built across existing communities and environments" (Bresnihan and Brodie, 2021). The governance gap is the recurring theme, captured in the observation from the Nordic case that "no one is steering this development, or closely looking at the impacts that it may have on remote communities in the Arctic and Nordic region" (Sovacool, Upham, and Monyei, 2022).

3.3 International evidence

Ireland supplies the clearest quantified case. Data centre demand there has risen from a small share to a dominant one, since “Data centre usage was only 5% in 2015” but “Data centres consumed 22% of Irish electricity in 2024” (The Irish Times, 2026). The consumer cost has been estimated directly, with analysis finding “Cumulatively for all Irish households: €715 million in price effects” and that “An estimated cumulative average of €360 has been added to household electricity bills over recent years” (The Irish Times, 2026). That cost coexists with substantial investment, as “Data centre investors have invested €18 billion into the Irish economy”, which is the trade-off the New South Wales inquiry must weigh rather than assume away (The Irish Times, 2026).

Virginia, the largest data centre market in the world, supplies the second case. A legislative review found that “unconstrained demand for power in Virginia would double within the next 10 years, with the data center industry being the main driver”, and that the cost falls broadly, since “data centers’ increased energy demand will likely increase system costs for all customers, including non-data center customers, for several reasons” (Joint Legislative Audit and Review Commission, 2024). The counterfactual matters, because “The state’s energy demand was essentially flat from 2006 to 2020 because, even though population increased, it was offset by energy efficiency improvements” (Joint Legislative Audit and Review Commission, 2024), which establishes that the new load, not background growth, is the driver. Australian analysts have already drawn the comparison, noting that “In Virginia, USA, unconstrained demand from data centres is projected to add almost $40 US to monthly residential electricity bills” (Energy Consumers Australia, 2025b).

3.4 The causal mechanism

The mechanism by which concentrated load can raise consumer bills is well characterised in the engineering and market literature, and it operates through three channels. The first is system scale, because a hyperscale facility is large relative to the grid that hosts it, and “With hyperscale datacenters exceeding 100 MW individually, and in some grids exceeding 15% of power load, DC adaptation is large enough to harm power grid dynamics, increasing carbon emissions, power prices, or reduce grid reliability” (Lin and Chien, 2023). The second is price formation, since the addition of large, continuous load affects “electricity markets and pricing, economic dispatch and reserve scheduling, and infrastructure planning and coordination” (Sheng et al., 2026), against a backdrop in which “The rapid growth of artificial intelligence (AI) is driving an unprecedented increase in the electricity demand of AI data centers, raising emerging challenges for electric power grids” (Chen et al., 2025). The third is cost recovery, because the network augmentation and system services that new load requires are recovered from consumers, and in Australia “electricity network costs make up almost half of household electricity bills” (Energy Consumers Australia, 2025a).

Whether these channels raise residential bills depends on how cost is allocated and on whether supply responds, so the mechanism is contingent rather than deterministic. The Australian market already applies a beneficiary-pays principle to some system services, since “the costs to provide FCAS are recovered from those who create the need, such as a large electricity user” (Energy Consumers Australia, 2025a). The price pressure is also conditional on the generation response, because the load is largely inflexible and facilities frequently draw carbon-intensive supply, since “datacenters often consume carbon-intensive energy from the grid when carbon-free energy is scarce” (Acun et al., 2023). The international price signal is nonetheless striking, with reporting that “wholesale electricity costs are as much as 267% higher today in areas near data centres than five years ago” and that, in Ireland, “data centres have been responsible for 88% of increased electricity demand from 2015 to 2024” (Energy Consumers Australia, 2025a).

3.5 The Western Sydney pipeline and grid outlook

The conditions that produced the effect overseas are present in Western Sydney in concentrated form. Clustering is the first condition. The State has approved what "The biggest data centre in the Southern Hemisphere has been given the green light by the Minns Labor Government", a facility of which "The facility is at Marsden Park in Sydney's north-west approximately 36 kilometres from the centre of Sydney" (NSW Government, 2025). Capacity concentration at the State level is acute, with New South Wales and Victoria together hosting "around 80% of the country's data centre capacity" (United States Studies Centre, 2025). Demand growth is the second condition. The New South Wales Government's own consultation paper relays that "AEMO forecasts that energy consumption from data centres will double from 5% of NSW's grid-supplied energy in 2026 to 11% by 2030" (Infrastructure NSW, 2026), a trajectory corroborated by the finding that data centres "Data centres currently account for 4% of the state's grid-supplied electricity in NSW, expected to rise to 11% by 2030" (United States Studies Centre, 2025). The committed pipeline alone is material, since "Data centres made up around 3 TWh of load in 2024, with the ESOO Central scenario forecasting around 5 TWh by 2033-34 from existing and committed projects alone" (Energy Consumers Australia, 2025b).

The network response is the third condition, and it is already being planned at scale. Transmission planning records that "Network Planning progressed studies to investigate capacity to support data centre load growth in Western Sydney", and that "Studies to-date indicate additional capacity of 2.8 GW to supply load can be enabled in western Sydney, if the following conditions are met" (Transgrid, 2024). Those conditions are extensive and are explicitly to be funded by the connecting parties, requiring that "Data centres connecting to the Transmission network fund the following augmentation projects: double-circuiting of line 14 between Kemps Creek and Sydney West with high-capacity conductors, approx. 280 MVAr dynamic voltage support, an additional 1,500 MVA 500/330 kV transformer at the Kemps Creek substation and data centre connection capacitor banks totalling 800 MVAr" (Transgrid, 2024), and further that "Committed BESS projects along Sydney's supply corridors totalling 2.6 GW proceed as planned" (Transgrid, 2024). The coordination risk is that these elements proceed out of step, because "Transmission investment decisions can lag data centre development timelines, and renewable projects may be delayed by network and supply challenges" (United States Studies Centre, 2025).

3.6 The distribution of cost

The decisive variable is who pays for that augmentation. Where connection costs are reflective and borne by the connecting party, the socialisation that drives gentrification is contained; where they are not, residential consumers absorb the difference through network charges that already "make up almost half of household electricity bills" (Energy Consumers Australia, 2025a). The New South Wales Government has articulated the correct principle, that "Data centre developers and operators need to fund their infrastructure requirements, in addition to what is already planned and funded, so as to not increase prices for households", and has proposed that "Cost recovery regimes should be reviewed at the respective national and state levels to ensure upgrades required for data centre connections are paid for by data centres" (Infrastructure NSW, 2026). That principle is currently a stated intention rather than a binding rule, and the evidence that load growth is already entering regulated revenue proposals is concrete, since "Jemena's revenue proposal for the 2026-2031 regulatory period outlined a significant increase in non-residential electricity consumption from 2024 levels due primarily to data centre load growth" (Energy Consumers Australia, 2025b). The wholesale dimension is equally contingent, with Australian modelling indicating that unmatched data centre growth could raise New South Wales wholesale prices by around 26 per cent by 2035, an impact that falls to around 3 per cent where matching renewable generation and storage proceed (Clean Energy Finance Corporation and Baringa, 2025).

3.7 Why no New South Wales finding yet exists

The absence of a completed New South Wales study reflects the recency of the load, not the absence of risk. The relevant research is being commissioned now, on the explicit premise that, "Based on international evidence, data centres could add hundreds of dollars annually to Australian household energy bills through a range of factors" (Energy Consumers Australia, 2025b). Until that work reports, the honest position is that the precursors are present and internationally evidenced, that the magnitude in New South Wales is not yet quantified, and that the outcome remains contingent on the cost-allocation and renewable-matching decisions the Committee and the Government are presently considering. Inferring a New South Wales figure from Irish or Virginian numbers would overstate the evidence, and this submission does not do so.

3.8 Recommendation

Protocol recommends a posture of monitoring with pre-emptive safeguards rather than retrospective remediation of the kind now under way in Ireland and Virginia. First, the consumer-bill research being scoped by Energy Consumers Australia should be commissioned with a Western Sydney regional breakdown and maintained as a standing dataset, so that the question can be answered with local evidence rather than overseas analogues (Energy Consumers Australia, 2025b). Second, the developer-funded principle already stated by the Government should be given binding legislative force, converting the intention that data centres "fund their infrastructure requirements" into an enforceable cost-recovery rule (Infrastructure NSW, 2026). Third, consent should carry minimum safeguards consistent with the Climate Council's submission, which recommends that proponents "Support demonstrably additional renewable generation and firming" and "Disclose and report consistent information on electricity use, water use, backup generation and emissions" (Climate Council of Australia, 2026). The renewable-matching requirement is the single most effective lever, because it is the variable that the modelling shows separates a 26 per cent wholesale impact from a 3 per cent one (Clean Energy Finance Corporation and Baringa, 2025), and because in its absence new load is liable to lean on existing or fossil supply built for other consumers (Greenpeace Australia Pacific, 2026). Finally, approvals, transmission planning and renewable zone development should be coordinated rather than run as "separate processes across federal and state jurisdictions" (United States Studies Centre, 2025).

Chapter 4

Synthesis and Recommendations

4.1 The common thread: measurement precedes allocation

The two questions appear distinct but reduce to one structural point. New South Wales is approving large, concentrated, largely inflexible load and contemplating efficiency conditions before it has the measurement and cost-allocation machinery that either task requires. An efficiency-conditioned regime in the Singaporean manner cannot allocate capacity against a Power Usage Effectiveness standard that is neither measured nor disclosed, and a consumer-protection posture cannot prevent energy gentrification without a binding rule on who funds augmentation. Both therefore turn on the same precondition, which the governance literature identifies as the reason compute is a tractable object of policy at all, namely that it is "detectable, excludable, and quantifiable" (Sastry et al., 2024). A recurring failure in comparable jurisdictions has been the converse, observed of Nordic datacentre development, where "no one is steering this development, or closely looking at the impacts that it may have on remote communities in the Arctic and Nordic region" (Sovacool, Upham, and Monyei, 2022). The remedy in both cases is to measure and disclose first, then allocate and condition.

4.2 Consolidated recommendations

(1) Make disclosure a condition of consent and continued operation for data centres above a defined information technology power threshold calibrated on the European Union's 500 kilowatt model, covering a measured basket of the NABERS Whole Facility rating, Power Usage Effectiveness, water usage effectiveness, absolute annual energy and renewable share, reported to the regulator and published (European Commission, 2024; Iman, 2025).

(2) Convert the Government's stated principle that data centres "fund their infrastructure requirements" into a binding cost-recovery rule, so that network and water augmentation driven by data centre connection is borne by the connecting party rather than socialised (Infrastructure NSW, 2026).

(3) Require demonstrably additional renewable generation and firming as a condition of consent, since this is the variable that separates a 26 per cent wholesale price impact from a 3 per cent one and prevents new load leaning on supply built for other consumers (Clean Energy Finance Corporation and Baringa, 2025; Climate Council of Australia, 2026; Greenpeace Australia Pacific, 2026).

(4) Coordinate approvals, transmission planning and renewable energy zone development, which are presently "undertaken through separate processes across federal and state jurisdictions" and consequently fall out of step (United States Studies Centre, 2025).

(5) Commission and maintain a standing dataset on consumer bill impacts and fleet efficiency, with a Western Sydney regional breakdown, so that policy is set against local measurement rather than overseas analogues (Energy Consumers Australia, 2025b).

(6) Require disclosure and limitation of backup and off-grid fossil generation, capturing the water and emissions dimensions that a Power Usage Effectiveness figure omits and that have historically gone unmeasured (Climate Council of Australia, 2026; Li et al., 2025).

These measures are mutually reinforcing. Disclosure makes the cost-recovery and renewable-matching rules auditable, the coordination measure prevents the timing failures that raise costs, and the standing dataset closes the evidentiary gap that currently prevents New South Wales from answering the Committee's second question with local numbers. The same instruments also create the option value of treating data centres as flexible grid assets rather than passive loads, an opportunity the literature documents where waste heat is valorised and "data centre operators in Stockholm and Paris direct waste heat through metropolitan district heating systems and urban homes" (Velkova, 2016).

4.3 Limitations

Three limitations should temper how this submission is read. First, several New South Wales figures derive from forecasts and from grey literature rather than from audited operational data, which is itself the strongest argument for the disclosure regime recommended above, since "the future impact of the data centers' electricity consumption is vulnerable to behavioral usage trends" and is therefore intrinsically uncertain (Koot and Wijnhoven, 2021). Second, Power Usage Effectiveness is a partial metric, and the water dimension in particular remains poorly measured even internationally, where "nearly half of servers are fully or partially powered by power plants located within water stressed regions" (Siddik, Shehabi, and Marston, 2021). Third, the energy gentrification conclusion is bounded by design, asserting a material and preventable risk rather than a demonstrated New South Wales effect, and it should be revisited when the consumer-impact research reports. The inquiry's own final report is the appropriate forum to consolidate that evidence, and the recommendations above are framed so that they can be adopted incrementally as it does.

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